EE 308 Lab 2

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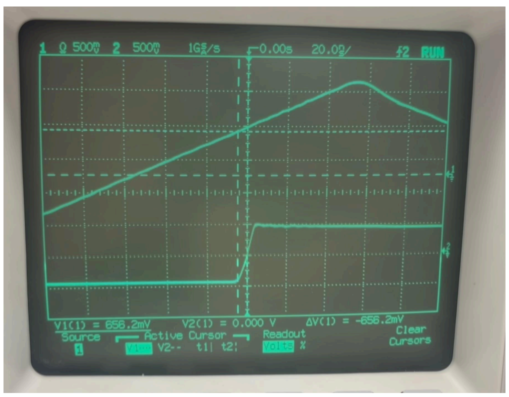
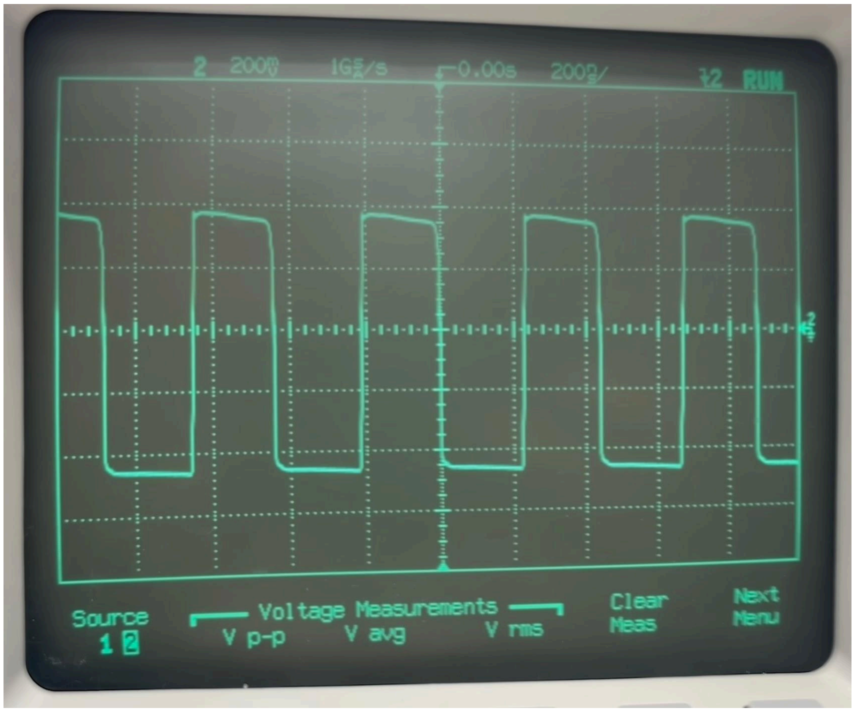
Specs Met

Input Resistance: We have a 1k series resistor to the base of our input differential pair. This was placed to meet the 1k input impedance requirement.

Trigger Thresholds: +0.5V/-0.5V with +/- 0.15V adjustability

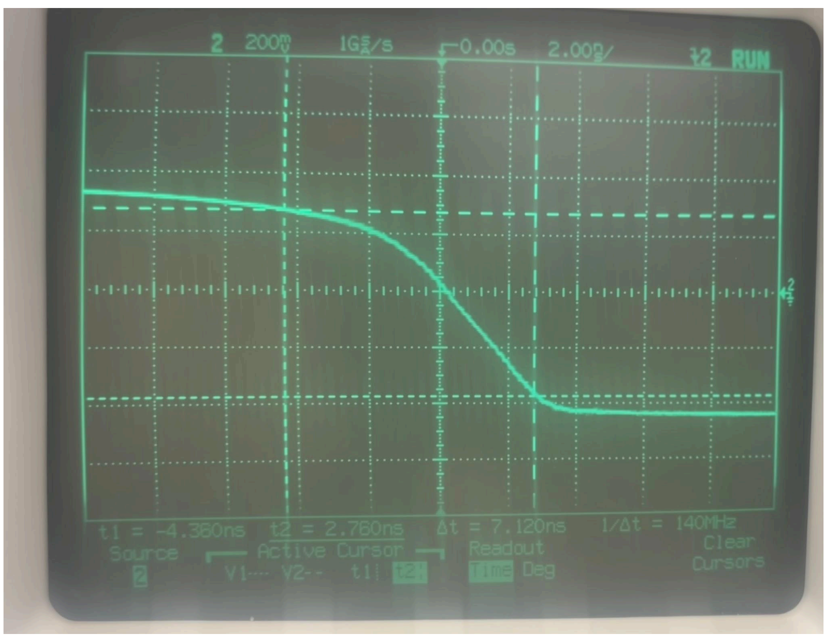
	Upper Threshold	**Lower Threshold**
Rise		
Fall		

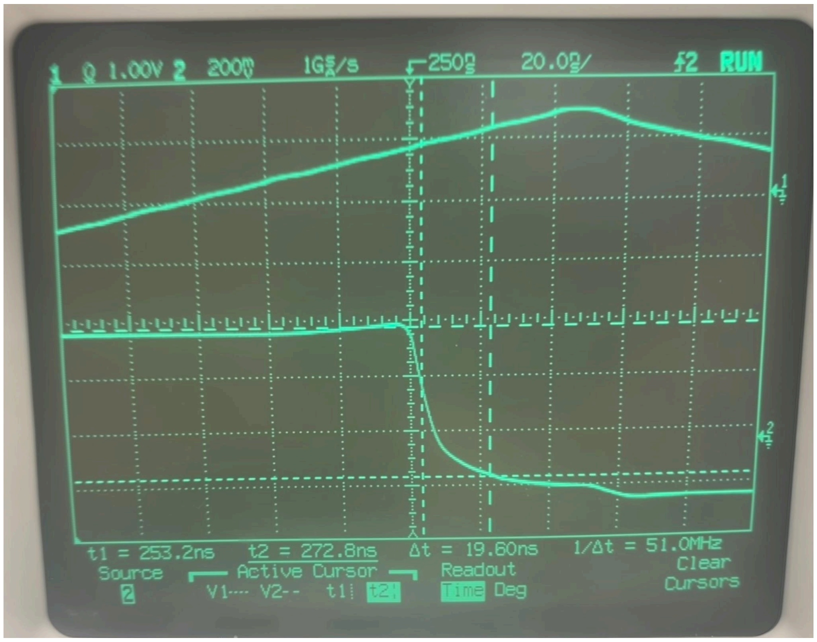
Square Wave Output: Our square wave output has a peak to peak voltage of 950mV.



Rise time: Our measured rise time is 6.64ns on output, on the feedback it is 9.8ns

Fall Time: Our measured fall time is 7.12ns on output, on the feedback 22.4ns





% Overshoot: From looking at the plot of multiple waveforms there is <2% overshoot.

Trigger Latency: From figuring out the exact triggering time at low frequency signal, and comparing it to the triggering time at 2MHz, there is 9.9ns of latency

2kHz	**2MHz**
	

Schematic

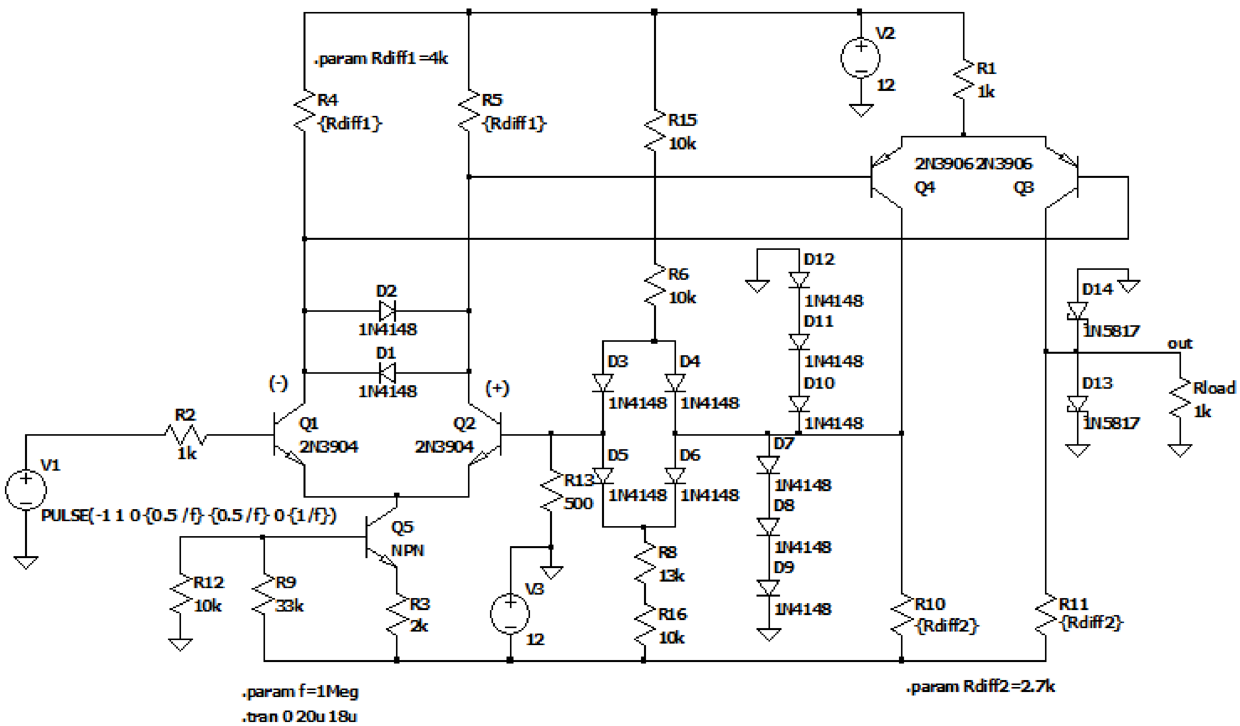
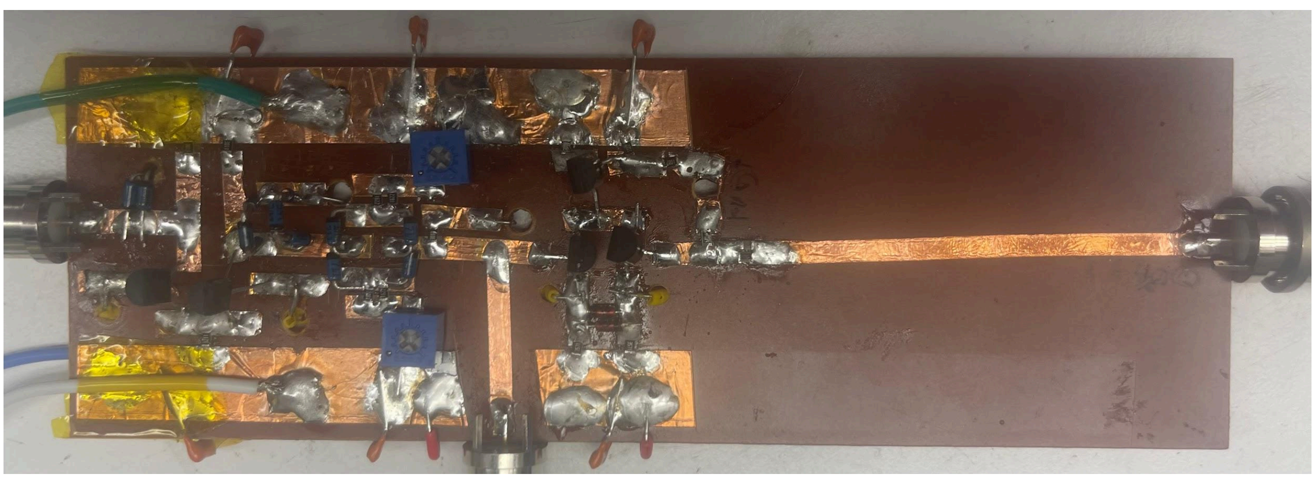


Photo of Board



Discussion

Our Schmitt trigger topology is based on the Schmitt trigger from the Tektronix CFG250 function generator.

A series 1k resistor is at the input of the Schmitt trigger to make the input high impedance. Beyond this resistor is the input of the Schmitt trigger which is a differential pair amplifier. The collectors are tied together with two parallel diodes to limit the output swing. This was done to speed up the differential pair since we don't need it to swing to and from its full saturation levels which ultimately take more time.

The tail of the first differential pair is a current source configured to sink roughly 4mA. The current source was configured by biasing the base of Q5 and using the diode drop from base to emitter to fix the voltage across R3. This then sets a fixed current of 4mA to be drawn from the tail of the input differential pair.

Both outputs of the first differential pair are then fed to another differential pair amplifier. One output is taken as the output of the entire Schmitt trigger to be fed to the output amplifier. The other output is fed back to the first differential pair amplifier. Its output is clamped to three diode drops to prevent large output swings that could potentially slow the Schmitt trigger down, as was done in the first differential pair. Before this clamped output is fed back it first passes through a diode bridge that sets the trigger thresholds. The diodes D3-D6 and resistors R6, R8, and R15-16 are configured such that only the parallel diodes will turn on. Thus when the output square wave is positive D3 and D6 conduct while D4 and D5 are off.

To find the values of the resistors we use the known thresholds and the swing range of the pot. The upper limit is 0.65V so with a 12V supply and a bottom resistor of 500 this makes the fixed upper resistor the following.

$$0.65 = 12 \cdot \frac{500\Omega}{500\Omega + R} \Rightarrow R \approx 8k\Omega$$

Then to find the potentiometer value we know that 0.35V will occur when the pot is fully on.

$$0.35 = 12 \cdot \frac{500\Omega}{500\Omega + 8k\Omega + R_{pot}} \Rightarrow R_{pot} \approx 10k\Omega$$

These values needed to be tuned slightly to get the final range. We also added D13 and D14 to clamp the output swing to roughly 1Vpp since we noticed some overshoot. We use Schottky diodes for this output node.